



Cape Peninsula
University of Technology

FACULTY OF INFORMATICS & DESIGN

DIPLOMA in INFORMATION & COMMUNICATIONS TECHNOLOGY

Subject Guide

APPLICATIONS DEVELOPMENT THEORY 3

SUBJECT GUIDE FOR:	2022
COURSE CODE:	ADT372S
NQF LEVEL:	7
PRE-REQUISITE SUBJECTS:	Applications Development Theory 2 Applications Development Fundamentals 2 Information Systems 2

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1. INTRODUCTION

Welcome to Applications Development Theory 3. This subject will build on the basic knowledge of developing applications acquired in the previous 2 years as well as incorporate database programming and software analysis and design concepts covered in Application Development Fundamentals 2 and Information Systems 2. A software engineering view will be taken in this subject for the development of applications. Applications Development Theory 3 will introduce students to the main concerns of software engineering practice and focus on methods, values, attitudes and techniques used. It will provide understanding of how to apply relevant methodological approaches to the development of well-designed and documented software solutions and the need for rigor during these processes.

2. GENERAL INFORMATION

2.1 Contact Information

DESIGNATION	NAME	LOCATION	TELEPHONE NUMBER	E-MAIL ADDRESS
Subject Lecturer (FT) and Course Coordinator	Ms G Khan	Engineering 2.8	(021) 460 3455	khang@cput.ac.za
Subject Lecturer (FT)	Ms F Allie	Engineering		allief@cput.ac.za
Subject Lecturer (PT)		As supplied in class and on Blackboard		
Head of Department	B. Kabaso	Engineering 2.30	(021) 460-3713	kabasob@cput.ac.za
Secretary	N. Allie	Engineering 2.1	(021) 460-3010	alien@cput.ac.za

2.2 Class Times and Venues

As published on <http://myclassroom.cput.ac.za> and notice boards. This subject consists of one lecture period per week. During term 1 there will also be one tutorial session per week.

2.3 Learner Management System (LMS)

Blackboard used for general communications and announcements, posting of course material, assignment and announcements. It is the responsibility of the learner to check Blackboard regularly at <https://myclassroom.cput.ac.za/>

2.4 Subject Credits

This subject has 20 credits at level 7 on the NQF. This means that the average student will take 200 hours per year to master the outcomes of the subject. Of this 30% of the time will be contact hours and the remainder will be self-study and other tasks, assignments and tests. Please refer to the Faculty prospectus on <http://www.cput.ac.za/academic/faculties/informaticsdesign/prospectus> for further information.

2.4.1 Length of Course

METHOD	NOTIONAL HOURS (1 Notional hour = 10 hours)
Formal Lectures	4
Practical / Assignments	5
Self-Study	10
Other	1
TOTAL	20

2.5 Ethical Conduct

Conduct in this subject is governed by CPUT's guidelines on the following ethical matters : -

TOPIC	CPUT REFERENCE DOCUMENT AVAILABLE AT
Roles and Responsibilities	CPUT's Policy Docs and Student Guidelines
	https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf
Class attendance and participation r	page 18 of CPUT's General Handbook Academic and Student Rules and Regulations
	https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf
Grievance procedures	page 61 of CPUT's General Handbook Academic and Student Rules and Regulations:
	https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf
Plagiarism	page 30 of CPUT's General Handbook Academic and Student Rules and Regulations:
	https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

*NB: An amended version of the General Handbook may be available on the university website.

2.6 Recognition of Prior Learning (RPL)

In a situation where there is a claim of previous learning that equates to a qualification of Information Technology, the recognition of such qualification will be at the discretion of the higher education institution. A structured process for the assessment of individual learners will be followed, vis-a-vis the exit level outcomes of the qualification on a case-by-case basis. Independent assessors will moderate this process and the assessment of individual cases.

In this course, we recognize RPL by granting a credit to learners who demonstrate both a theoretical and practical knowledge of software engineering and develop large software intensive systems. This is not to be confused with an exemption granted for a similar subject studied at another recognized tertiary institution.

3. COURSE INFORMATION

3.1 Overview of the Course

The need for automation in order to promote efficiency and effectiveness of human operations in diverse fields of application cannot be over-emphasized. The concept of software engineering enables the design and development of software intensive systems based on sound engineering principles. This course is designed to provide understanding of some of the modern concepts of software engineering and their application to solving industrial scale problems. There will provide an opportunity for students to work together in groups that model software project teams in order to deliver some project objectives based on assigned tasks.

3.2 Course Objectives

- 3.2.1 To provide students with an understanding of modern software engineering concepts and their application to industrial scale problems;
- 3.2.2 To provide students with the knowledge of software design and modelling as core activities of the software development process;
- 3.2.3 To study a myriad of tools and technologies that are relevant to different software process models and soft implementation approaches; and
- 3.2.4 To enable students with the capability to design and develop usable software intensive systems based on the application of modern software engineering concepts, tools and techniques.

At the end of this course and having completed the Essential reading and activities, you should be able to:

- describe the expectations, pressures and problems faced in developing software and the need for processes, tools, techniques and approaches
- outline the underlying processes of software engineering and critically assess relevant approaches
- analyse, design, test and maintain software systems and document these actions correctly

3.3. Lecture Delivery Methods

The mode of lecture delivery will be based on the following:

- Online Lectures, tutorials
- Mandatory Reading of the prescribed textbook. Topics must be read prior to being covered in class
- Software Engineering Assignments (Group and individual)

3.4. Textbooks

The prescribed textbook which is essential to this course is listed below. This book has been selected since a free PDF copy can be downloaded along with the PowerPoint slides for each chapter. There are also videos available for most of the chapters on YouTube and these are narrated by the author.

Software Engineering, by Ian Sommerville either the 9th or 10th edition, Pearson Education Limited.

3.5. Library and Readings

Additional reading is an essential part of the course and you are expected to make use of the library and the Internet

4. ASSESSMENT

4.1 Assessment policy and regulations

This subject has four assessments. Assessments are a combination of Blackboard online tests, iLearning quizzes and assignments. Assignments are administered via Blackboard.

Test queries: Last day for any queries regarding a test is one week after the test was discussed in class. Thereafter the test mark cannot be changed.

Absence from a test: If you are absent from a test, a valid medical certificate must be given to your lecturer (via the Secretary) within one week after the test was written. You must also inform your lecturer on the day of the test that you are ill.

Due dates must be strictly adhered to. NO late submissions are allowed for any subject in this Diploma.

4.2 Assessment Opportunities:

There will be one test per term. At the end of the year there will be an Exception/Sick test for the semester.

	ASSESSMENT BREAKDOWN AND WEIGHTS						
	1st SEMESTER			2nd SEMESTER			
ASSESSMENT CRITERIA	1st TERM	2nd TERM		3rd TERM	4th TERM		TOTAL
Nature of assessment	Assignments	Assignments	Test	Assignments	Assignments	Test	
Will assessment be included in this term?	Yes	Yes	Yes	Yes	Yes	Yes	
Total weight for the assessment	5%	15%	30%	10%	10%	30%	100%
Will the assessment be moderated?	No	No	Yes	No	No	Yes	

4.3 Tests and Assignment Guidelines

For a group work the maximum number of students per group will vary. Please consult the Assessment Guide for details for each assignment. Group work should be done in a way that encourages individual contributions. Deadlines for submission of assignments must be adhered to in order to qualify for the maximum score.

4.3.1 Assignments

- Assignments on software design and development will target specific learning outcomes
- Students will work in distinct teams to undertake all assigned classwork
- the classwork will entail working with specific case studies by applying learned concepts from lectures, and some assigned academic readings in order to design and implement software systems

5 SUBJECT INFORMATION

5.1 Subject Outcomes

At the end of this course, students will be able to:

OUTCOME NO	OUTCOME DESCRIPTION
SO1	Software Engineering processes, Agile Software Development and Requirements Engineering
SO2	Modelling, Architectures and Testing
SO3	Software Quality Management and Evolution
SO4	Sociotechnical systems and Use

5.2 Critical Cross-Field Outcomes (CCFO's)

The Critical Cross-Field Outcomes to be addressed by this course are problem solving, group work, organization and management, application, evaluation, communication, presentation, and research.

Students will have the opportunity to enhance their problem solving skills through the acquisition of theoretical concepts that underline database design, and application of the knowledge gained to solve the real world problems. Group work and communication are addressed by collaborative work on solving the database problems.

5.3 Topic Organisation

Term 1 THEME 1: SOFTWARE ENGINEERING PROCESSES AND PRACTICE		
No	Topic	Week starting
1	Introduction to Software Development and Software Engineering	31/1/2022
2	Software processes for developing software and engineering software	7/2/2022
3	Agile Software Engineering	14/2/2022
4	Requirements Engineering	28/2/2022
Term 1 Assignments on Requirements Engineering		
Term 2 THEME 2: MODELLING, ARCHITECTURES and TESTING		
No	Topic	Week starting
5	Systems Modelling	28/3/2022
6	Architectural Design	11/4/2022
7	Design and Implementation	25/4/2022
8	Testing	2/5/2022
Term 2 Assignments and Exam on Semester 1 work		
Term 3 THEME 3: SOFTWARE QUALITY MANAGEMENT AND EVOLUTION		
No	Topic	Week starting
9	Software Evolution	18/7/2022
10	Sociotechnical Systems	18/8/2022
Term assignment Assignments on Software Evolution and Sociotechnical Systems		
Term 4 THEME 4: SOCIOTECHNICAL SYSTEMS AND USE		
No	Topic	Week starting
11	Software Dependability	19/9/2022
	Assignment 4	
Term 4 assignment and Exam on Semester 2 work		

6. STUDENT ACKNOWLEDGEMENT

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY STUDENT ACKNOWLEDGEMENT

2022

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ACKNOWLEDGE THAT YOU HAVE READ THE SUBJECT GUIDE AND UNDERSTAND ITS CONTENT

Please read the following statement and complete the spaces that follow. Make a copy of this page for your own reference. Hand-in this original page during the next class session.

I have read the subject guide of **APPLICATIONS DEVELOPMENT THEORY 3** and understand the information and my responsibilities as a learner for this subject. I am also aware that the Subject Guide may be revised due to unforeseen circumstances and it remains my responsibility to adhere to the most recent guide.

Name & Surname:
(Print)

Student Number:

Contact number: (home/res) Contact number: (cell)

Signature:

Date: